Candlestick geometry and market regimes: a quantitative approach

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* Outspoken Market, March 2025
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Abstract – Identifying market regimes is a central problem in quantitative finance. Nevertheless, many traditional approaches rely on univariate descriptors such as returns and volatility, or on latent-state models whose historical classification may change under re-estimation. This paper proposes an alternative, observable measure of regime based on the local geometry of OHLC candles. Specifically, the method computes the determinant of the covariance matrix of normalised open, high, low, and close values over rolling windows, interprets this determinant as a proxy for the geometric volume of price structure, and transforms it through logarithmic scaling and a rolling percentile rank to classify the market into compressed, neutral, and expanding states. The empirical analysis focuses on the S&P500 index contract (US500) at daily, 4-hour, and 1-hour frequencies, complemented by exploratory evidence across other assets. The results suggest that the indicator preserves qualitative coherence across timeframes while taking on distinct analytical roles at each scale, from a structural background in the daily horizon to more tactical, execution-sensitive behaviour at shorter frequencies. The proposed measure is not intended to forecast price direction directly, but to serve as a state variable describing the structural environment in which price movements occur. Its main contribution is to transform candle geometry into a simple, observable, and economically interpretable quantitative proxy for regime, with potential applications in filtering, model conditioning, regime-switching systems, and multi-timeframe quantitative frameworks.

Keywords – *market regime identification; OHLC geometry; covariance matrix determinant; log-determinant; state variables; quantitative finance.*

I. Introduction

In quantitative finance, market regime identification is a central problem. Strategies, predictive models, and risk management systems often depend on distinguishing between environments characterised by compression, expansion, trend, transition, or structural instability. In practice, it is rarely enough to forecast the next return. It is also necessary to understand the type of environment in which that return is generated. The same signal can behave very differently across market regimes (Hamilton, 1989; Ang & Timmermann, 2012).

The literature and applied practice offer several approaches to this problem. These include volatility-based measures, trend filters, range estimators, temporal clustering techniques, and regime-switching models such as Markov chains and Hidden Markov Models (Hamilton, 1989; Rabiner, 1989). These approaches are useful and, in many cases, capture important aspects of market dynamics. However, they also have notable limitations. Latent-state models rely on statistical inference about structures that are not directly observable. As a result, regime classification can be sensitive to model specification, sample size, and the arrival of new data. Operationally, this implies that interpretations of the past may be revised retrospectively: states previously classified in one way may be reassigned after re-estimation, reducing the historical stability of the labelling (Hamilton, 1989; Ang & Timmermann, 2012).

This point is especially important when the goal is not merely to fit the data well but to build state variables that can be used in applied research, real-time monitoring, and integration into quantitative models. In such settings, there is value in regime measures that depend less on latent inference and are more directly grounded in the observable properties of the series itself. The proposal of this paper fits precisely within that direction. Rather than inferring the regime from a hidden probabilistic structure, it is measured through the local geometry of candles, that is, through the way open, high, low, and close jointly organise over time (Ang & Timmermann, 2012).

This change in perspective matters because a large share of traditional descriptors in quantitative finance relies on one-dimensional series, especially returns and volatility (Campbell, Lo and MacKinlay, 1997; Engle, 1982; Bollerslev, 1986). This choice is natural, as returns tend to be more statistically tractable than price levels, and volatility plays a central role in risk description. Still, this reduction often discards structural information. A candle is not merely a close, nor simply a range. It has an internal configuration expressed by the relationships among open, high, low, and close. This configuration captures information on dispersion, asymmetry, range extension,

intraday persistence, and the relative intensity of local price movements. In other words, the candle has a geometric shape (Parkinson, 1980; Garman & Klass, 1980).

The central hypothesis of this paper is that different market regimes produce distinct geometries in OHLC space, and that these geometries can be summarised operationally by a simple multivariate measure: the determinant of the covariance matrix computed over a rolling window of normalised candles. From a mathematical standpoint, the determinant of a covariance matrix measures the volume of the dispersion ellipsoid associated with the data (Johnson & Wichern, 2007; Mardia, Kent and Bibby, 1979). When this volume is small, the joint structure of the variables tends to collapse along redundant directions, suggesting compression, local linearity, or low geometric richness. As volume increases, dispersion occurs in multiple directions, suggesting greater structural complexity and a more active regime.

Applied to the OHLC context, this interpretation offers a conceptually distinct alternative to traditional regime approaches. Rather than relying exclusively on return magnitude, univariate risk proxies (Engle, 1982; Bollerslev, 1986), or ex-post-estimated latent states, the method uses a directly observable property of the local price structure. In addition, by using the log-determinant and contextualising it with historical percentiles, the approach turns this quantity into a state variable comparable over time, enabling a simple, operational classification into compressed, neutral, or expanding regimes.

The motivation, therefore, is not to deny the usefulness of HMMs or classic regime-switching systems (Hamilton, 1989; Rabiner, 1989), but to offer a methodological counterpoint: a structural, local, observable, and retrospectively more stable measure that can serve as a standalone market-reading tool and as an input to broader models. In this formulation, regime is treated not as a hidden state to be inferred but as a geometric property to be measured.

Building on this idea, the present work proposes a conceptual and methodological interpretation of the determinant of the OHLC covariance matrix as a measure of geometric regime in quantitative finance. To the best of our knowledge, the contribution of this work is not to introduce new mathematical elements in isolation, but to propose a new articulation among them: applying the determinant of the covariance matrix of normalised OHLC series as an observable proxy for geometric market regime. The goal is not to forecast future price direction directly, but to construct a variable capable of identifying states of compression, neutrality, and structural expansion, while preserving economic interpretability, operational simplicity, and integration potential with broader quantitative applications.

II. Methodology

The proposed methodology aims to construct an observable measure of structural regime from the local geometry of candlesticks. To do so, it uses joint information on open, high, low, and close, treating it as a multivariate time series. The central idea is that different market regimes differ not only in return magnitude or volatility intensity, but also in how the four OHLC dimensions jointly organise across successive observation windows (Campbell, Lo and MacKinlay, 1997; Engle, 1982; Bollerslev, 1986).

The data used in this study consist of OHLC time series obtained via TradingView, chronologically ordered and analysed at daily, 4-hour, and 1-hour frequencies, using the S&P500 index contract as the main empirical object. The choice of asset and frequency defines the empirical context of the exercise but does not alter the method's general logic. Although the code includes auxiliary variables for contemporaneous and forward returns, these quantities are not central to the methodological proposal. The study focuses on constructing a geometric state variable rather than a supervised directional forecasting model. In addition, an exploratory analysis was conducted across other assets in different market classes, with the objective of qualitatively assessing the conceptual stability and portability of the proposed measure beyond the study's main asset.

The first step normalises the four OHLC dimensions using a simple moving average of the close, with a window length of P = 66. Formally, for each time t, the normalised variables are defined as $open_t \ / \ MA_t$, $high_t \ / \ MA_t$, $low_t \ / \ MA_t$, and $close_t \ / \ MA_t$, where MA_t is the moving average of the close over the 66-bar window ending at t. Next, a constant scale factor, here equal to 1000, is applied only to improve numerical stability and computational readability, without changing the relative structure of the data. This normalisation is important because it reduces the influence of the absolute price level and allows the analysis to focus on the candle's relative shape rather than its nominal scale.

From this normalised base, for each time t greater than or equal to the window length, an observation matrix X_t of dimension P x 4 is constructed. Each row of this matrix represents a normalised candle within the rolling window, and each column corresponds to one OHLC dimension. On this matrix, a $4x4$ sample covariance matrix is computed (Johnson & Wichern, 2007; Mardia, Kent and Bibby, 1979), summarising the dispersion and co-movement structure among open, high, low, and close over the window. Because these variables are strongly related, the matrix may become numerically close to singular during periods of high structural redundancy. To avoid instability, a very small regularisation term is added to the diagonal, defined as epsilon = 10^-12 times the identity matrix. This correction does not change the economic interpretation of the indicator but improves computational robustness.

The next step is to compute, for each window, the determinant of the regularised covariance matrix. In geometric terms, this determinant can be interpreted as the volume of the dispersion ellipsoid generated by observations in OHLC space. When the value is very low, it suggests that the four dimensions are moving in a highly redundant, compressed manner, or are close to a lower-dimensional subspace (Johnson & Wichern, 2007; Mardia, Kent, & Bibby, 1979). When the value increases, dispersion occurs across multiple directions, indicating a richer, less degenerate structure, potentially associated with greater local complexity in the price formation process. Thus, the determinant becomes more than an algebraic statistic and functions as a measure of the geometric market regime.

Because the covariance determinant can vary across several orders of magnitude, the methodology uses its logarithm as the primary operational variable. The log-determinant improves scale stability, reduces distributional skewness, and facilitates temporal analysis. In practical terms, it makes the measure more interpretable, both visually and for subsequent quantitative applications. The log transform also aligns with the fact that the interest lies not in the raw magnitude of geometric volume but in its relative variation over time.

However, the log-determinant alone is not sufficient to define regimes in a way that is comparable through time. Therefore, the method adds a historical contextualization step through a rolling percentile rank. At each time point, the percentile position of the current log-determinant is computed relative to the previous 252 values of the series. In this way, regime classification depends not on a fixed absolute level, but on the relative position of the current geometry within the asset's recent history and the analysed frequency. This choice is important because the same log-determinant value may represent compression in one context and expansion in another. The rolling percentile rank solves this problem by making the reading adaptive and relative to the local statistical environment of the series.

Based on this rolling percentile, a simple operational taxonomy of three regimes is defined. Values at or below the 20th percentile are classified as a compressed regime, values at or above the 80th percentile are classified as an expanding regime, and intermediate values are classified as a neutral regime. This classification does not claim to capture a definitive ontology of the market but rather offers a practical convention for separating states into low, medium, and high geometric complexity. In other words, regime here is not treated as a latent state inferred through optimisation, but as an observable, relative, and updatable property of OHLC structure.

Finally, the methodology organises the information into three complementary panels: closing price, the log-determinant series, and the rolling percentile rank with the regime classification. Moreover, periods classified as compressed or expanding are visually highlighted on the price chart, allowing inspection of whether changes in local geometry coincide with relevant transitions in market dynamics. This visual step is not merely illustrative. It serves as an exploratory verification of the indicator's usefulness, enabling examination of whether compression blocks precede expansions, whether geometric extremes accompany phases of stronger displacement, and whether the proposed variable exhibits enough persistence to be treated as an operational measure of structural regime.

In summary, the methodology transforms candle geometry into a quantitative state variable through a simple and interpretable sequence of steps: OHLC normalisation, rolling covariance estimation, determinant computation, logarithmic transformation, and rolling percentile-based classification. The result is an observable regime proxy that aims to capture not only how much the market moves, but how its internal structure organises over time.

III. Results, analysis, and business interpretation

The results indicate that the determinant of the OHLC covariance matrix, when logarithmically transformed and contextualised by a rolling historical percentile, produces a coherent reading of geometric regime across multiple time scales. This coherence does not imply identical signals across frequencies, but rather conceptual stability of the method: the same mathematical object appears to capture, in each horizon, a specific dimension of market structure. Across all analysed frequencies, daily, 4-hour, and 1-hour, the indicator alternates between compressed, neutral, and expanding states, with persistence sufficient for economic interpretation and potential operational use as a state variable.

In the daily chart, the indicator behaves as a structural background measure. Expansion blocks appear in periods when the log-determinant rises into the upper range of its recent distribution, suggesting that daily candles occupy a larger volume in OHLC space. Economically, this is consistent with phases of greater complexity in the price-formation process, in which the relationships among open, high, low, and close become richer and less redundant. This behaviour can emerge under stronger displacement, flow acceleration, persistent range expansion, or higher intraday heterogeneity. Conversely, compression blocks in the daily chart suggest a relative collapse in geometric complexity. This may be associated with both consolidation phases and cleaner trend periods, in which the daily candle preserves direction but loses internal structural richness. At this scale, the indicator ceases to be merely a short-term oscillator and becomes a descriptor of the market's background structure.

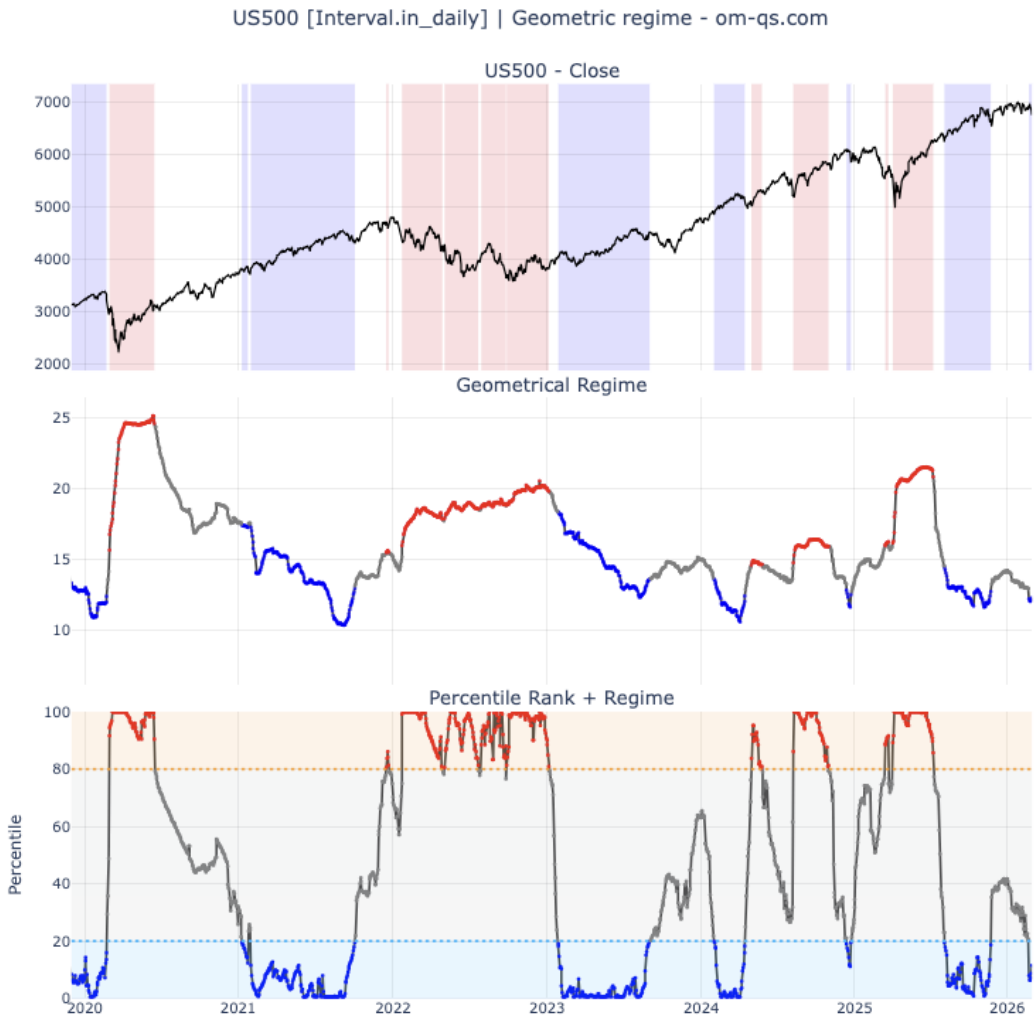

Figure 1. Geometric regime in the S&P500 futures contract (US500) at the daily frequency. The top panel shows the price with visual markers for blocks classified as compressed and expanding; the middle panel shows the log-determinant of the rolling OHLC covariance matrix; and the bottom panel shows the rolling historical percentile used for regime classification.

At the 4-hour frequency, the reading becomes especially interesting because the indicator maintains sensitivity to relevant changes without the same level of noise as at the hourly frequency. Compression and expansion blocks appear more organised, with durations sufficient for tactical reading and transitions clearer than in the 1-hour chart. This suggests that the 4-hour scale can operate as an intermediate layer between structure and execution. Economically, it captures moments when the market moves from a state of low effective dimensionality to a phase of higher joint dispersion across OHLC variables, even before these changes fully consolidate in the daily chart. For this reason, the 4-hour horizon appears particularly useful as a transition or tactical regime, that is, a scale that can signal environmental change without reacting excessively to local fluctuations.

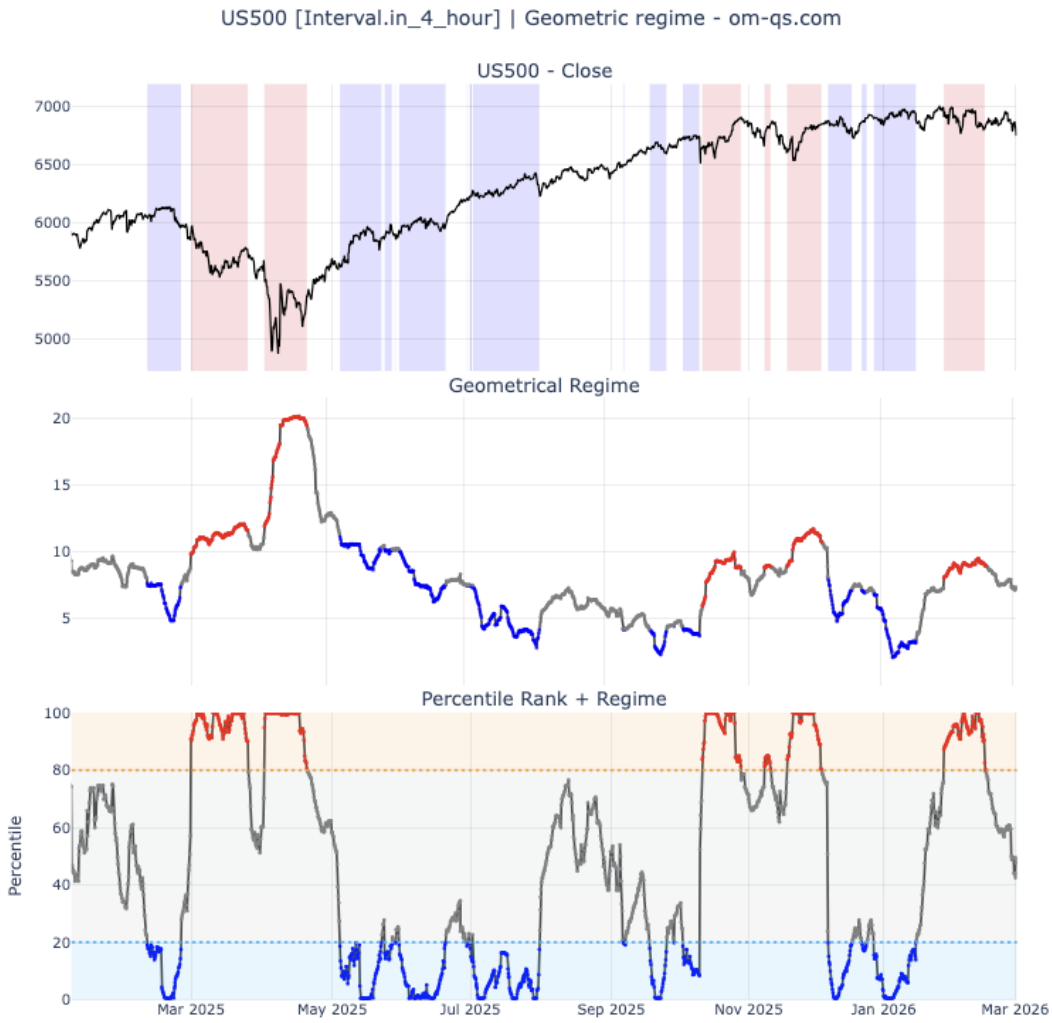

Figure 2. Geometric regime in US500 at the 4-hour frequency. The panels have the same interpretation as Figure 1.

In the 1-hour chart, the indicator exhibits the most sensitive behaviour and the closest alignment with short-horizon microstructure. Alternations between compression and expansion occur more frequently, and the percentile moves more rapidly between the extreme bands. This is consistent with the nature of the frequency: over short horizons, small changes in candle shape, intraday range, and the relationships among open, high, low, and close are already sufficient to alter the local geometry of the rolling window materially. Economically, expanding periods at this scale suggest moments when the market's microstructure is more "filled", with greater variety in intraday behaviour. In contrast, compressed periods reflect a simplification of candle shapes and higher redundancy among their dimensions. Therefore, the 1-hour horizon seems less suitable as a master structural regime, but potentially very useful as an execution or timing layer within a multi-timeframe architecture.

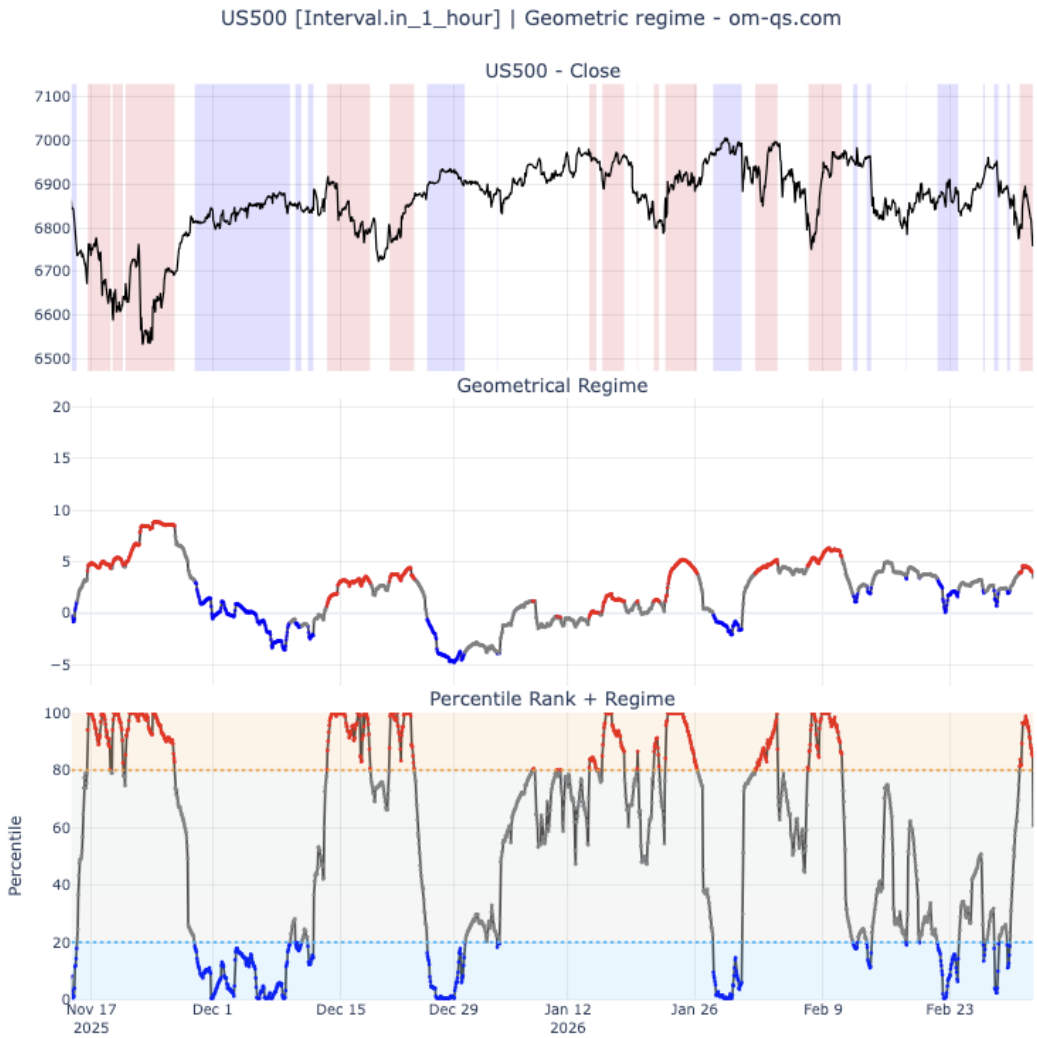

Figure 3. Geometric regime in US500 at the 1-hour frequency. The panels have the same interpretation as Figure 1.

One of the most relevant points in the charts is that the indicator appears to capture something distinct from price direction. Across the three frequencies, the market can rise, fall, or move sideways while the geometric regime alternates between compression, neutrality, and expansion. This supports the methodological interpretation that the percentile-ranked log-determinant is not a directional signal but a state variable (Ang & Timmermann, 2012). Its role is not, by itself, to anticipate whether the next move will be up or down, but to describe the structural environment in which that move occurs. This distinction is particularly valuable in quantitative finance because it separates the regime-filtering problem from the directional forecasting problem, rather than forcing a single variable to perform both functions simultaneously.

Another important finding is cross-frequency compatibility. The three charts suggest a plausible time hierarchy for the practical use of the indicator. The daily chart offers a slower, broader structural regime reading. The 4-hour chart organises transitions into more usable tactical blocks. The 1-hour chart provides the fastest, most sensitive, and most execution-adjacent reading. This compatibility is among the most promising results, as it suggests that the concept of geometric regime is not lost when the operational horizon changes. Instead, it appears to adapt to scale while preserving the same mathematical logic and changing its analytical role. In practical terms, this opens the door to multi-timeframe systems in which the daily horizon serves as a master regime, the 4-hour horizon as a tactical filter, and the 1-hour horizon as an execution layer.

The general economic interpretation of the results is that high log-determinant values, particularly in the upper percentiles of the recent distribution, are consistent with a structurally richer market, less redundant and with greater local complexity. Low values in the lower percentiles are consistent with low effective dimensionality in OHLC behaviour, that is, an environment where candle shapes become simpler, more confined, and less informationally diverse. This reading does not replace volatility, ATR, or other traditional descriptors, but provides an additional interpretive layer. Rather than measuring only how much the market moves, the indicator aims to measure how its internal geometry organises over time (Engle, 1982; Bollerslev, 1986; Parkinson, 1980; Garman & Klass, 1980).

Overall, the results suggest that the OHLC covariance determinant is a promising proxy for a structural market regime. Its main strength is transforming candle geometry into a state variable that is comparable over time, has a plausible economic interpretation, is coherent across frequencies, and is amenable to integration within broader quantitative systems. The method does not diminish the usefulness of classic regime approaches but offers a relevant counterpoint: rather than inferring hidden states subject to retrospective revisions, it directly measures an observable property of local price structure.

Although the main analysis in this work focuses on the S&P500, an exploratory evaluation was also conducted across other markets, including the Brazilian equity index futures (WIN1!), a volatility index (VIX), the Nasdaq equity index (US100), gold (GOLD), and Bitcoin (BTCUSDT). The objective of this step is not to provide exhaustive statistical validation across assets, but to verify whether the conceptual logic of geometric regime remains economically interpretable when applied to market classes with distinct microstructures, volatility profiles, and dynamics.

In the case of WIN1!, the indicator shows relatively clear compression and expansion blocks over time, suggesting it can capture relevant structural changes in Brazilian equity index futures. In several segments, expanding regimes coincide with phases of stronger price dynamism, while compressed periods occur in environments of lower relative geometric richness. This reinforces the idea that the method does not depend exclusively on the FX market structure and can be applied to directional assets sensitive to domestic risk flows.

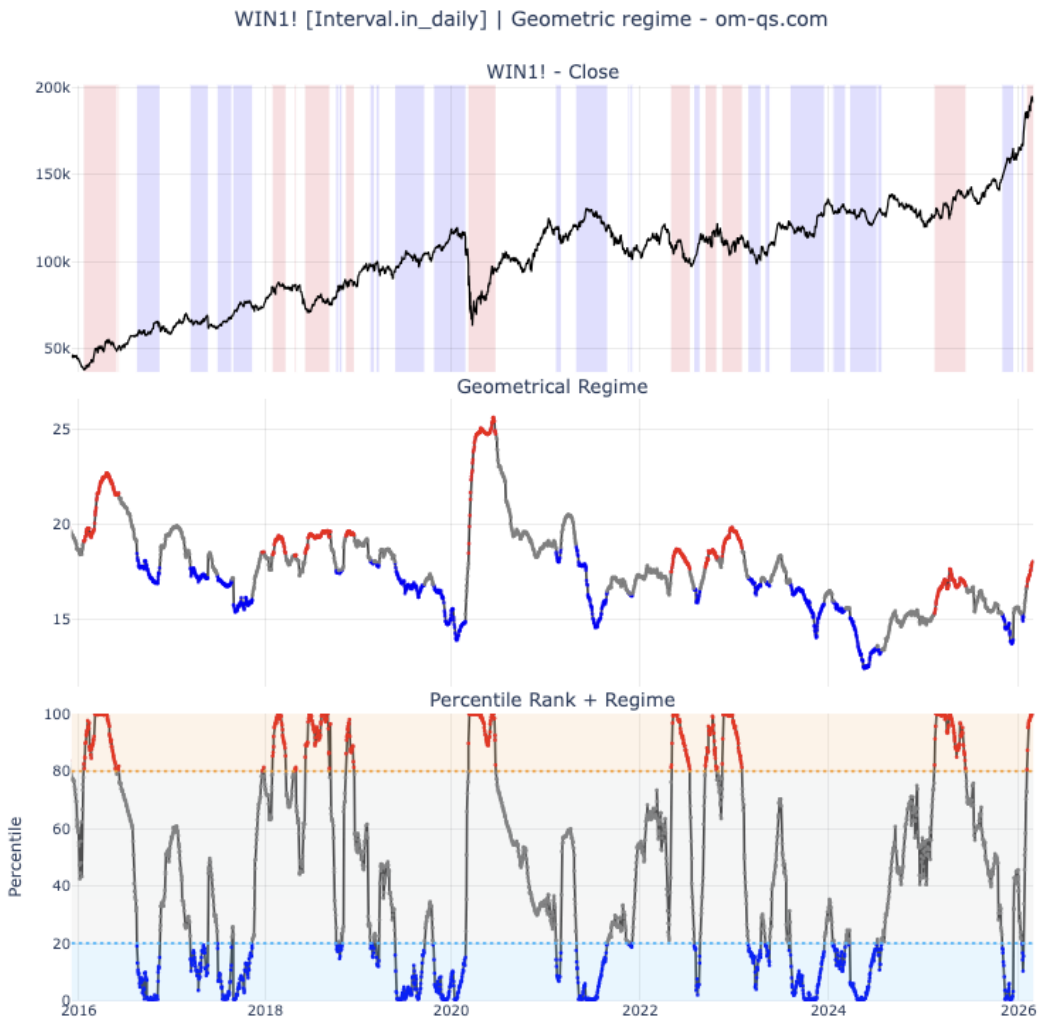

Figure 4. Geometric regime indicator applied to WIN1! at the daily frequency.

In VIX, the reading becomes particularly interesting because the asset is intrinsically linked to risk perception and the market's implied convexity. Still, the indicator does not reduce to the index level but appears to capture changes in the local geometry of its temporal

formation. Observed expanding blocks suggest moments of higher structural complexity, possibly associated with stress phases, transitions, or sharper repricing of implied volatility. This is relevant because it indicates that the proposed measure is not merely reflecting a price trend but an internal organisation of the process.

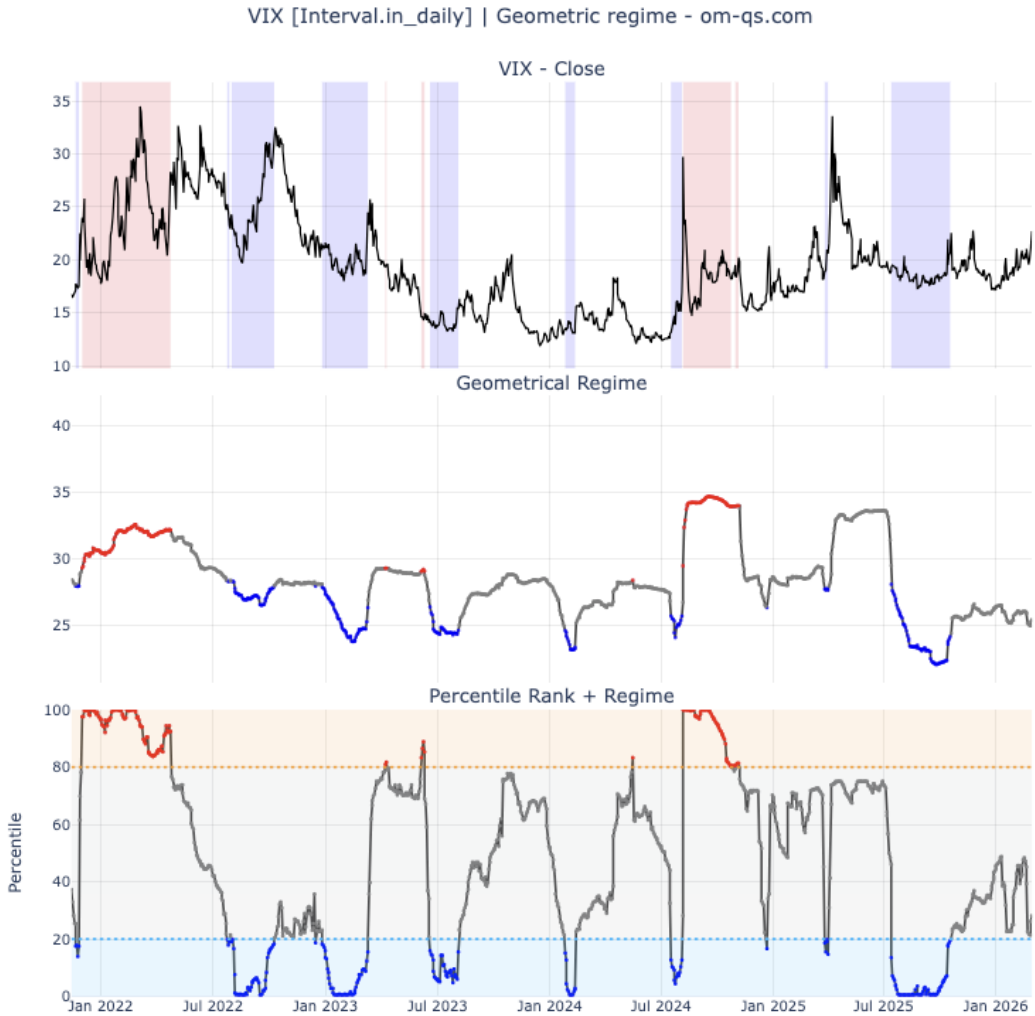

Figure 5. The geometric regime indicator is applied to VIX at the daily frequency.

For the Nasdaq, the geometric regime shows good readability in a broad, liquid market that is structurally distinct from the S&P500 environment. Alternation between compression and expansion is observed across very different cycles, including during recovery phases and periods of directional continuation. This suggests that the indicator can extract structural information even in broad indices, where the relative smoothness of trends could otherwise create a false impression of lower local complexity. Here again, the measure behaves more like a state variable than a simple reflection of market direction.

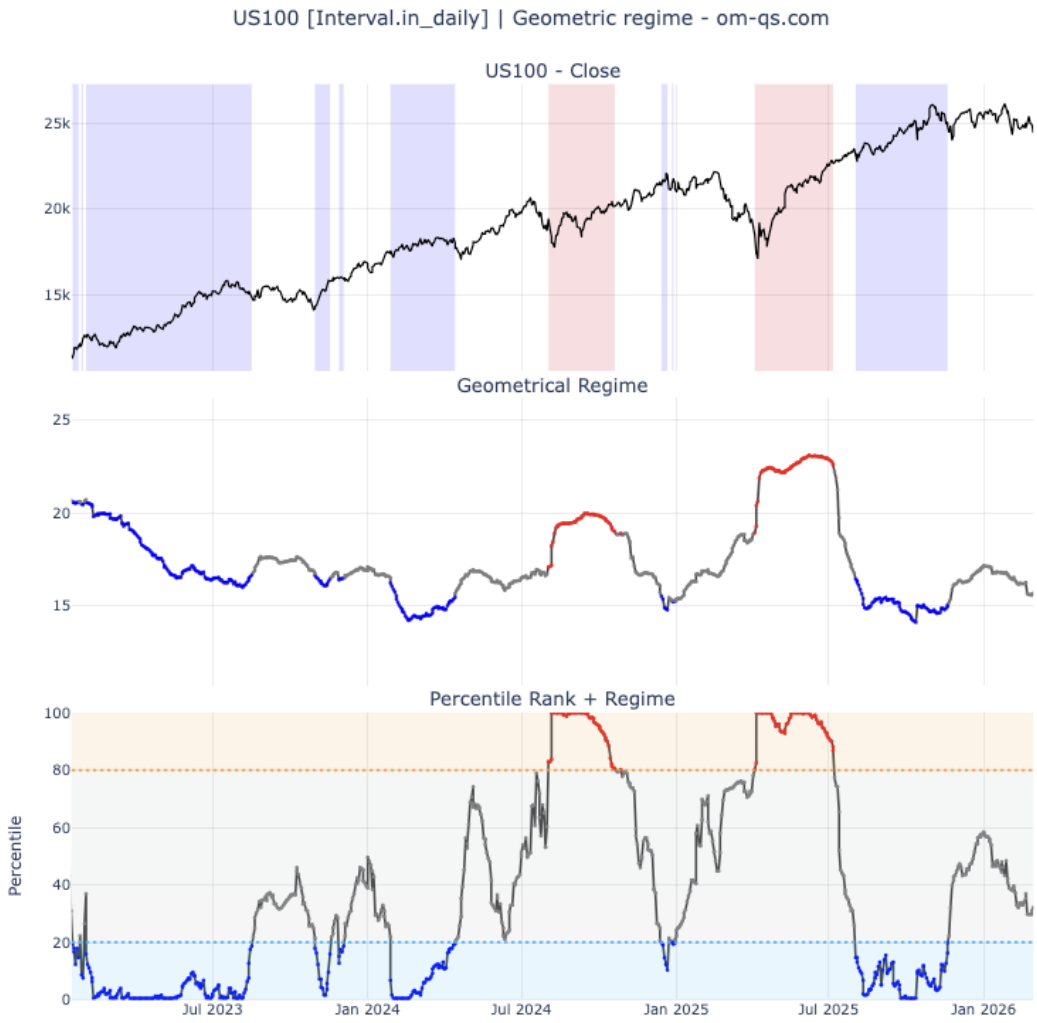

Figure 6. Geometric regime indicator applied to US100 at the daily frequency.

In gold (GOLD), the indicator also shows coherent behaviour, with expansion episodes associated with phases of higher structural intensity in price movement. Because gold often alternates between long trend periods and accommodation phases, the presence of distinct regime blocks suggests that OHLC geometry can differentiate moments of higher and lower internal candle richness, even in an asset traditionally viewed as a store of value and a macro hedge. This broadens the method's relevance beyond local equities and derivatives.

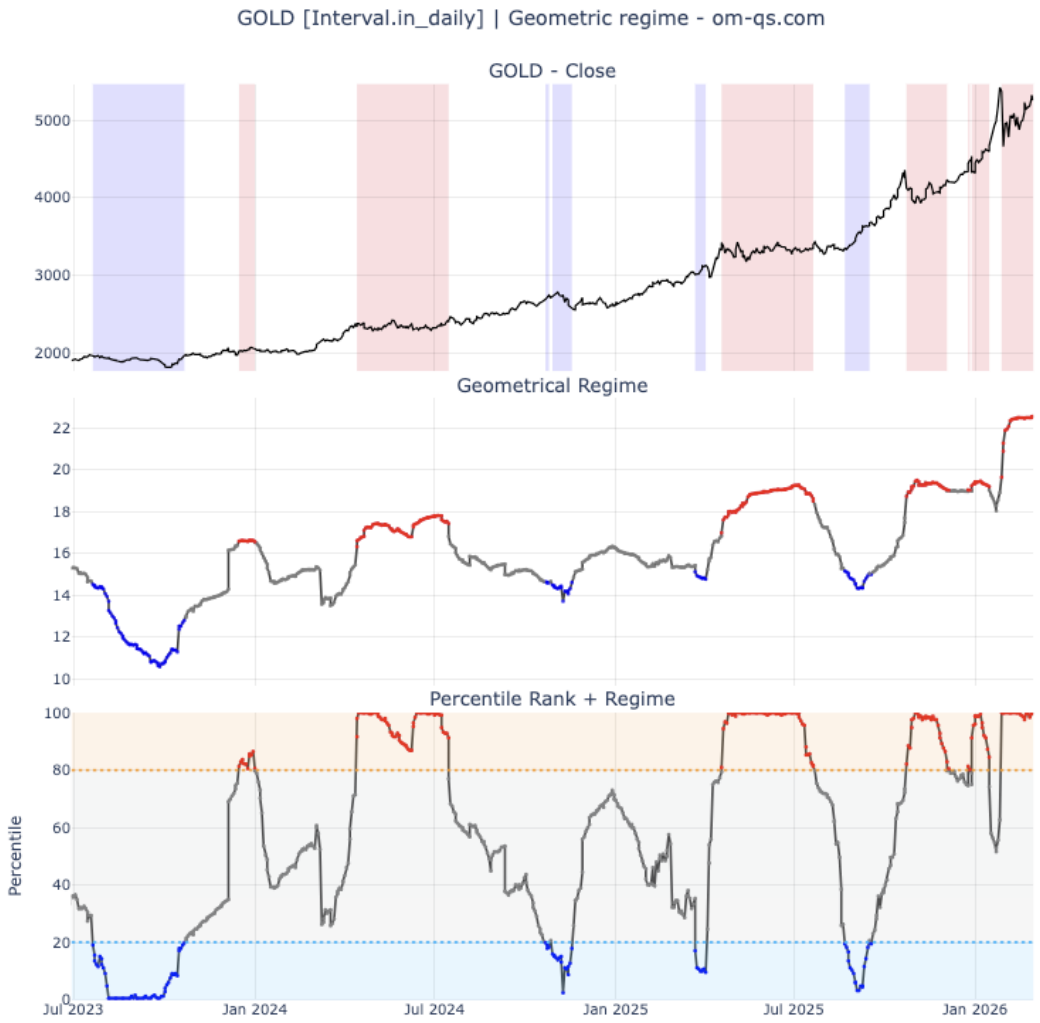

Figure 7. Geometric regime indicator applied to GOLD at the daily frequency.

In BTCUSDT, the application is particularly useful given the asset's well-known volatility and pronounced behavioural heterogeneity over time. The indicator shows well-defined compression and expansion blocks, with relatively clear shifts in the rolling percentile. This result is important because it suggests that the measure can also operate in noisier markets with structures distinct from those of traditional assets. The application to Bitcoin therefore reinforces the hypothesis that geometric regime is not restricted to conventional markets but may provide a more general reading of local price structure.

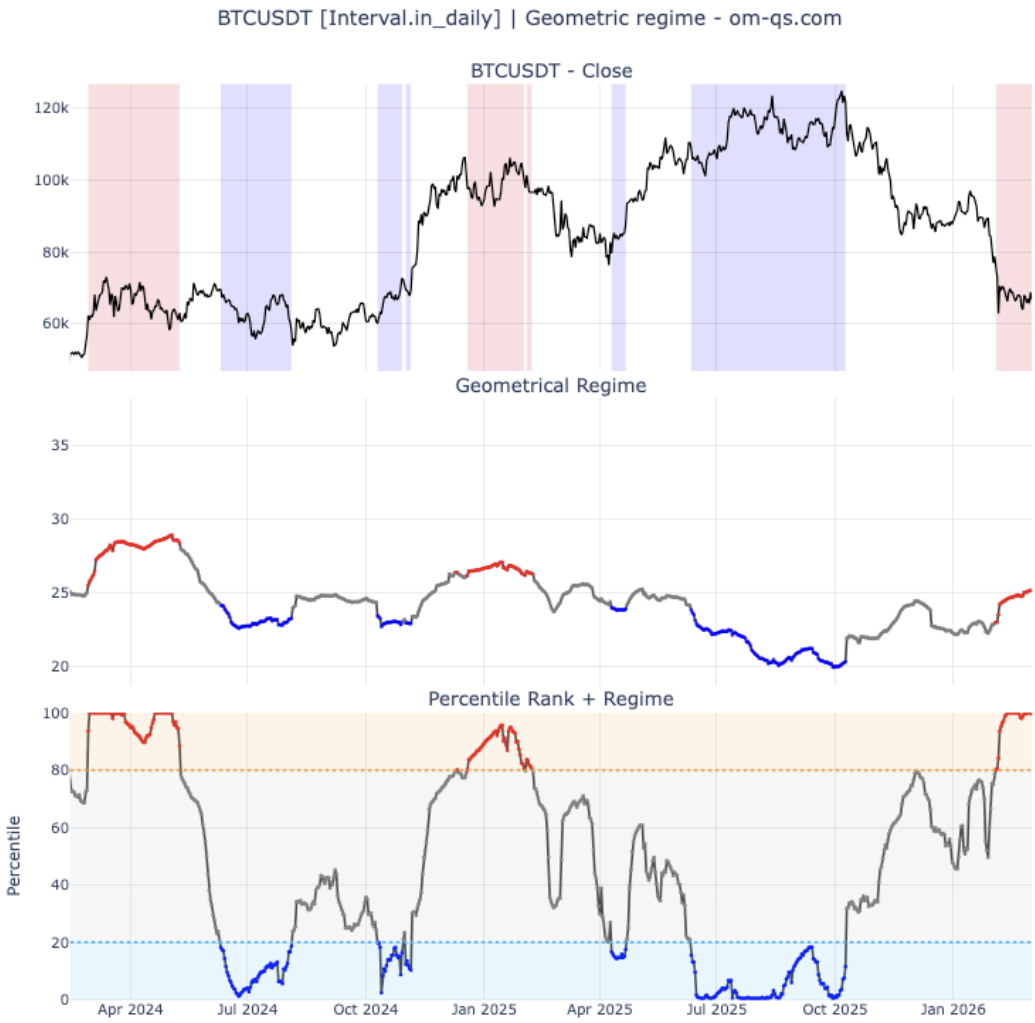

Figure 8. Geometric regime indicator applied to BTCUSDT at the daily frequency.

Overall, the results suggest that the indicator retains its qualitative coherence beyond the main asset. In all cases, alternation between compressed, neutral, and expanding states is observed, with persistence sufficient for visual reading and economic interpretation. At the same time, each asset seems to express this geometry in a distinct way, consistent with its liquidity characteristics, trading structure, event sensitivity, and candlestick patterns.

IV. Conclusion and suggestions for future work

This paper proposed a new way to measure market regimes from the local geometry of candlesticks, using the determinant of the covariance matrix of OHLC variables as a structural proxy. The central idea was to treat regime not as a latent state inferred by probabilistic models subject to retrospective re-estimation, but as an observable property of the joint organisation of open, high, low, and close over time.

Results for the SP&500 futures contract at daily, 4-hour, and 1-hour frequencies suggest that this approach can produce a coherent and economically interpretable reading of geometric regime. In particular, the indicator distinguished contexts of compression, neutrality, and structural expansion, while preserving sufficient persistence for analytical reading and possible operational application. In addition, the comparison across time scales indicates that the measure maintains conceptual stability across frequencies while assuming distinct functions at each horizon: more structural on the daily chart, more tactical on the 4-hour chart, and more sensitive to micro-dynamics on the 1-hour chart.

From a methodological standpoint, the main contribution is the introduction of a multivariate, simple, and directly observable measure that complements traditional regime approaches. Rather than depending exclusively on returns, volatility, or hidden states, the method seeks to capture a candle's internal shape as a quantitative information object. This shift enables the incorporation of a geometric dimension into the regime problem, a dimension that, although often present in the data, is often neglected by conventional one-dimensional descriptors.

At the same time, the results should be interpreted with caution. The study is exploratory in nature and does not claim that the indicator, by itself, exhausts the description of market regime or fully replaces methods already consolidated in the literature. Nor does it argue that the measure is directly directional. Its role, as shown throughout the paper, is to act as a state variable, that is, a descriptor of local market structure rather than an isolated predictor of the next price move.

Still, the findings suggest that the OHLC covariance determinant is a promising proxy for studying regimes in quantitative finance. Its computational simplicity, geometric interpretability, and adaptability to multiple frequencies make it particularly attractive for applied research and integration into broader quantitative systems. The additional exploratory analysis across other assets also suggests qualitative portability across market classes, reinforcing its potential as a broadly usable state variable in quantitative finance.

As a natural continuation of this work, future research may pursue several directions. These include formal statistical validation across different assets and market classes; direct comparison with classic regime-switching models such as Hidden Markov Models and Markov Switching Models; investigation of incremental value when combined with traditional measures of volatility, trend, or microstructure; and development of multi-timeframe architectures in which the geometric regime acts as a hierarchical filter for allocation, model selection, or execution.

In summary, this study suggests that candle geometry contains relevant structural information about the market and that this information can be transformed into a simple, operational quantitative state variable. By shifting attention from price direction to the joint OHLC shape, the proposal offers a useful counterpoint to latent regime approaches and opens a promising research agenda in quantitative finance.

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